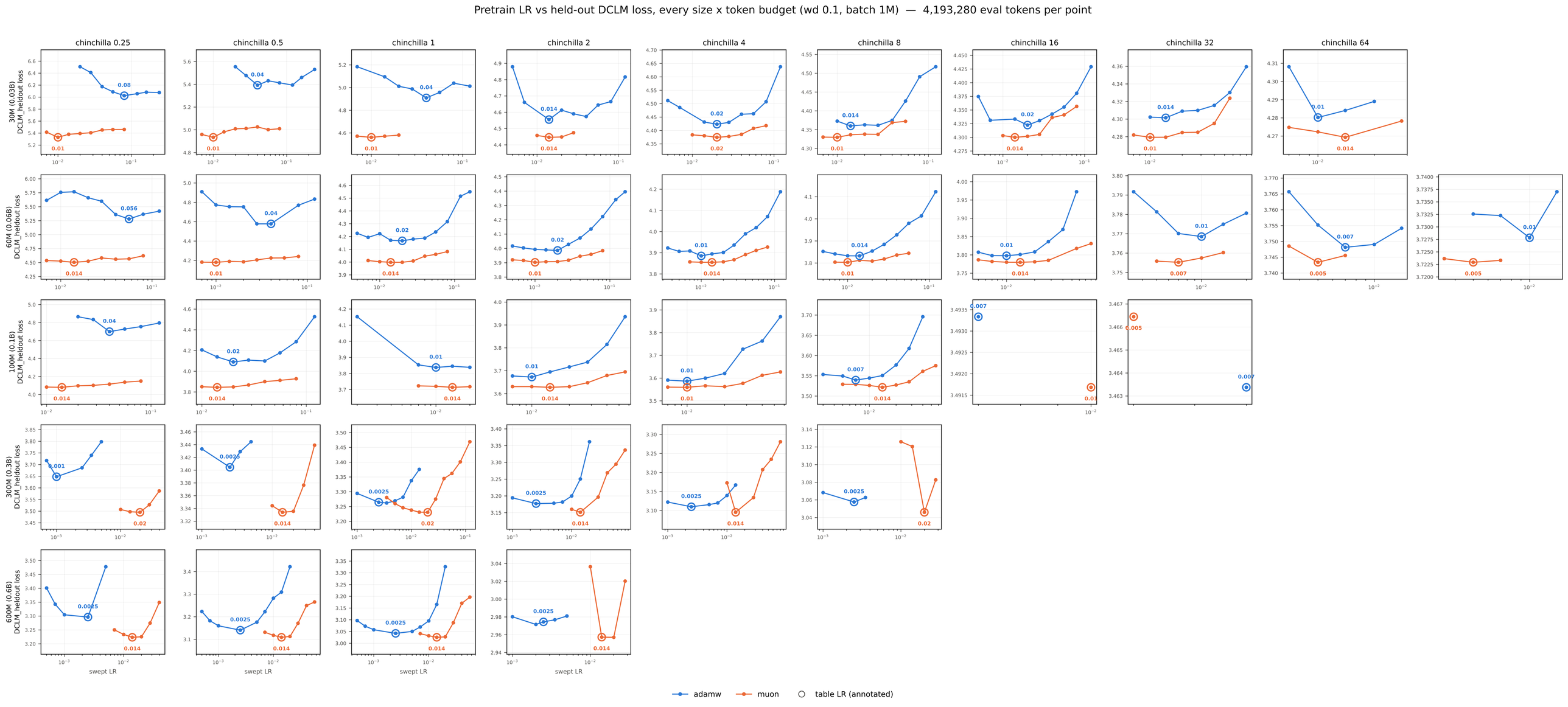

Pretrain LR vs held-out DCLM loss, every size x token budget (wd 0.1, batch 1M) — 4,193,280 eval tokens per point
chinchilla 0.25
chinchilla 0.5
chinchilla 1
chinchilla 2
chinchilla 4
chinchilla 8
chinchilla 16
chinchilla 32
chinchilla 64
30M (0.03B) DCLM_heldout loss
60M (0.06B) DCLM_heldout loss
100M (0.1B) DCLM_heldout loss
300M (0.3B) DCLM_heldout loss
600M (0.6B) DCLM_heldout loss
swept LR
adamw
muon
table LR (annotated)